

A Probabilistic Comparison of Commonly Used Interest Measures for Association Rules

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Abstract

This document contains a comprehensive collection of commonly used measures of significance and interestingness (sometimes also called strength) for association rules and itemsets. Interest measures are usually defined in terms of itemset support and counts. Here, we also present their relationship with estimating probabilities and conditional probabilities.

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An annotated bibliography of association rules can be found at https://mhahsler.github.io/arules/docs/association_rules.html.

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Code and Implementation

All measures discussed on this page are implemented in function `interestMeasure()` in the freely available software:

- R package: `arules`
- Python module: `arulespy`

Corrections and Feedback

For corrections and missing measures on this page or in the implementation in the package `arules`, please open an issue on GitHub or contact me directly.

Definitions

Agrawal, Imielinski, and Swami (1993) define association rule mining in the following way:

Let $I = \{i_1, i_2, \dots, i_m\}$ be a set of m binary attributes called **items**. Let $D = \{t_1, t_2, \dots, t_n\}$ be a set of transactions called the **database**. Each transaction $t \in D$ has a unique transaction ID and contains a subset of the items in I , i.e., $t \subseteq I$. A **rule** is defined as an implication of the form $X \Rightarrow Y$ where $X, Y \subseteq I$ and $X \cap Y = \emptyset$. The sets of items (or **itemsets**) X and Y are called the antecedent (left-hand side or LHS) and consequent (right-hand side or RHS) of the rule, respectively. Measures of importance (interest) can be defined for itemsets and rules. The support-confidence framework defines the measures support and confidence. Rules that satisfy user-specified minimum thresholds on support and confidence are called **association rules**.

Interest measures are usually defined in terms of itemset support. Here, we also present them using probabilities and, where appropriate, counts. The probability $P(E_X)$ of the event that all items in itemset X are contained in an arbitrarily chosen transaction can be estimated from a database D using maximum likelihood estimation (MLE) by

$$\hat{P}(E_X) = \frac{|\{t \in D; X \subseteq t\}|}{n}$$

where $n_X = |\{t \in D; X \subseteq t\}|$ is the number of transactions that contain the itemset X , and $n = |D|$ is the size (number of transactions) of the database. For conciseness of notation, we will drop the hat and the E from the notation for probabilities. We will use in the following $P(X)$ to mean $\hat{P}(E_X)$ and $P(X \cap Y)$ to mean $\hat{P}(E_X \cap E_Y) = \hat{P}(E_{X \cup Y})$, the probability of the intersection of the events E_X and E_Y —that is, the probability that a transaction contains all items in the union of the itemsets X and Y . The event notation should not be confused with the set notation used in measures like support, where $\text{supp}(X \cup Y)$ means the support of the union of the itemsets X and Y .

Note on probability estimation: These probability estimates can be very poor for itemsets with low observed frequencies. This limitation affects most measures discussed below and should always be considered.

Note on null transactions: Transaction data sets typically contain a large number of transactions that do not contain either X or Y . These transactions are called null transactions, and it is desirable that measures of rule strength are not influenced by a change in the number of null transactions. However, most measures are affected by the number of null transactions since the total number of transactions is used for probability estimation. Measures that are not influenced by a change in the number of null transactions are called null-invariant (Tan, Kumar, and Srivastava 2004; Wu, Chen, and Han 2010).

Good overview articles about different association rule measures are

- Tan, Kumar, and Srivastava (2004) Selecting the right objective measure for association analysis. *Information Systems*, 29(4):293-313, 2004
- Geng and Hamilton (2006) Interestingness measures for data mining: A survey. *ACM Computing Surveys*, 38(3):9, 2006.
- Lenca et al. (2007) Association Rule Interestingness Measures: Experimental and Theoretical Studies. *Studies in Computational Intelligence (SCI)* 43, 51–76, 2007.

Measures Defined on Itemsets

Support

Reference: Agrawal, Imielinski, and Swami (1993)

$$supp(X) = \frac{n_X}{n} = P(X)$$

Support is defined on itemsets and gives the proportion of transactions that contain X . It is used as a measure of significance (importance) of an itemset. Since it uses the count of transactions, it is often called a **frequency constraint**. An itemset with support at least as large as a specified minimum support threshold, $supp(X) \geq \sigma$, is called a **frequent or large itemset**.

For a rule, support is defined as the support of all items in the rule, i.e., $supp(X \Rightarrow Y) = supp(X \cup Y) = P(X \cap Y)$.

Support has the **downward-closure property (anti-monotonicity)**, which means that all subsets of a frequent set are also frequent. This property (actually, the fact that no superset of an infrequent set can be frequent) is used to prune the search space (usually thought of as a lattice or tree of itemsets with increasing size) in level-wise algorithms (e.g., the Apriori algorithm).

The disadvantage of support is the **rare item problem**. Items that occur very infrequently in the data set are pruned, although they would still produce interesting and potentially valuable rules. The rare item problem is important for transaction data, which usually have a very uneven distribution of support across items (typically a power-law distribution in which a few items occur frequently and most occur rarely).

Range: $[0, 1]$

Support Count

Alias: Absolute Support Count

Range: $\{0, 1, \dots, n\}$, where n is the number of transactions.

All-Confidence

Reference: Omiecinski (2003)

All-confidence is defined on itemsets (not rules) as

$$\text{all-confidence}(X) = \frac{supp(X)}{\max_{x \in X} supp(\{x\})} = \frac{P(X)}{\max_{x \in X} P(\{x\})}$$

where $\max_{x \in X} supp(\{x\})$ is the support of the item with the highest support in X . All-confidence means that all rules that can be generated from itemset X have at least a confidence of $\text{all-confidence}(X)$. All-confidence possesses the downward-closure property and thus can be effectively used inside mining algorithms. All-confidence is null-invariant.

Range: $[0, 1]$

Cross-Support Ratio

Reference: Xiong, Tan, and Kumar (2003)

Defined on itemsets as the ratio of the support of the least frequent item to the support of the most frequent item, i.e.,

$$\text{cross-support}(X) = \frac{\min_{x \in X} supp(\{x\})}{\max_{x \in X} supp(\{x\})}$$

An itemset is a cross-support pattern if this ratio is smaller than a specified threshold. Such patterns combine frequent and rare items and are often spurious.

Range: $[0, 1]$

Measures Defined on Rules

Contingency Table

A 2×2 contingency table with counts for rule $X \Rightarrow Y$ in the transaction dataset. The counts are:

	Y	$\bar{Y}$	Total
X	n_{XY}	$n_{X\bar{Y}}$	n_X
$\bar{X}$	$n_{\bar{X}Y}$	$n_{\bar{X}\bar{Y}}$	$n_{\bar{X}}$
Total	n_Y	$n_{\bar{Y}}$	n

n_{11} is the number of transactions that contain all items in X and Y . The first subscript indicates whether X is present and the second indicates whether Y is present. All other measures for rules can be calculated using these counts.

Confidence

Alias: Strength

Reference: Agrawal, Imielinski, and Swami (1993)

$$conf(X \Rightarrow Y) = \frac{supp(X \Rightarrow Y)}{supp(X)} = \frac{supp(X \cup Y)}{supp(X)} = \frac{n_{XY}}{n_X} = \frac{P(X \cap Y)}{P(X)} = P(Y|X)$$

Confidence is defined as the proportion of transactions that contain Y among those that contain X . This proportion estimates the probability of observing the rule's consequent given that a transaction contains the antecedent.

Confidence is directed and gives different values for the rules $X \Rightarrow Y$ and $Y \Rightarrow X$. Association rules must satisfy a minimum confidence constraint, $conf(X \Rightarrow Y) \geq \gamma$.

Confidence is not downward closed and was developed together with support by Agrawal et al. (the so-called support-confidence framework). Support is first used to find frequent (significant) itemsets exploiting its downward closure property to prune the search space. Then confidence is used in a second step to produce rules from the frequent itemsets that exceed a minimum confidence threshold.

A limitation of confidence is its sensitivity to the frequency of the consequent Y . Consequents with higher support tend to produce higher confidence values even when the items are independent.

Range: $[0, 1]$

Added Value

Alias: AV, Pavillon Index, Centered Confidence

Reference: Tan, Kumar, and Srivastava (2004)

Added value quantifies how much the probability of Y changes after conditioning on transactions that contain X . It is defined as

$$AV(X \Rightarrow Y) = conf(X \Rightarrow Y) - supp(Y) = P(Y|X) - P(Y)$$

Range: $[-1 + 1/n, 1 - 1/n]$ for an empirical data set with n transactions (0 indicates independence)

Causal Confidence

Reference: Kodratoff (2001)

Causal confidence combines a rule's confidence with the confidence of its contrapositive:

$$\text{causal-conf} = \frac{1}{2}[conf(X \Rightarrow Y) + conf(\bar{Y} \Rightarrow \bar{X})] = \frac{1}{2}[P(Y|X) + P(\bar{X}|\bar{Y})]$$

Range: $[0, 1]$

Causal Support

Reference: Kodratoff (2001)

Causal support adds the proportion of examples to the proportion of contrapositive examples:

$$\text{causal-sup} = supp(X \cup Y) + supp(\bar{X} \cup \bar{Y}) = P(X \cap Y) + P(\bar{X} \cap \bar{Y})$$

Range: $[0, 1]$

Centered Confidence

Alias: relative accuracy, gain

Reference: Lavrač, Flach, and Zupan (1999)

$$CC(X \Rightarrow Y) = conf(X \Rightarrow Y) - supp(Y)$$

Range: $[-1 + 1/n, 1 - 1/n]$

Certainty Factor

Alias: CF, Loevinger

Reference: Galiano et al. (2002)

The certainty factor measures the change in the probability of Y when conditioning on X , scaled by the maximum possible positive change. Positive values indicate positive association, and negative values indicate negative association.

$$CF(X \Rightarrow Y) = \frac{conf(X \Rightarrow Y) - supp(Y)}{supp(\bar{Y})} = \frac{P(Y|X) - P(Y)}{1 - P(Y)}$$

Range: $(-\infty, 1]$ for this positive-change normalization (0 indicates independence)

Chi-Squared

Reference: Brin, Motwani, and Silverstein (1997)

For the analysis of 2×2 contingency tables, the chi-squared test statistic is a measure of the relationship between two binary variables (X and Y). It can be used to test independence between X and Y :

$$\begin{aligned} \text{chi-squared}(X \Rightarrow Y) &= \sum_i \frac{(O_i - E_i)^2}{E_i} \\ &= \frac{(n_{XY} - \frac{n_X n_Y}{n})^2}{\frac{n_X n_Y}{n}} + \frac{(n_{\bar{X}Y} - \frac{n_{\bar{X}} n_Y}{n})^2}{\frac{n_{\bar{X}} n_Y}{n}} + \frac{(n_{X\bar{Y}} - \frac{n_X n_{\bar{Y}}}{n})^2}{\frac{n_X n_{\bar{Y}}}{n}} + \frac{(n_{\bar{X}\bar{Y}} - \frac{n_{\bar{X}} n_{\bar{Y}}}{n})^2}{\frac{n_{\bar{X}} n_{\bar{Y}}}{n}} \\ &= n \frac{[P(X \cap Y)P(\bar{X} \cap \bar{Y}) - P(X \cap \bar{Y})P(\bar{X} \cap Y)]^2}{P(X)P(Y)P(\bar{X})P(\bar{Y})} \end{aligned}$$

O_i is the observed count of contingency table cell i and E_i is the expected count given the marginals. The statistic has approximately a χ^2 distribution with 1 degree of freedom for a 2×2 contingency table. The critical value for $\alpha = 0.05$ is 3.84; higher chi-squared values indicate that the null hypothesis of independence between the LHS and RHS should be rejected. Larger values indicate stronger evidence of an association. The statistic can be converted into a p-value using the χ^2 distribution.

Notes: The contingency tables for some rules may contain cells with low expected values (less than 5), so Fisher's exact test might be more appropriate. Each rule represents a statistical test, and correction for multiple comparisons may be necessary.

Range: $[0, \infty)$

Collective Strength

Reference: Aggarwal and Yu (1998)

$$\begin{aligned} S(X, Y) &= \frac{1 - v(X, Y)}{1 - E[v(X, Y)]} \frac{E[v(X, Y)]}{v(X, Y)} \\ &= \frac{P(X \cap Y) + P(\bar{X} \cap \bar{Y})}{P(X)P(Y) + P(\bar{X})P(\bar{Y})} \frac{P(X)P(\bar{Y}) + P(\bar{X})P(Y)}{P(X \cap \bar{Y}) + P(\bar{X} \cap Y)}. \end{aligned}$$

where $v(X, Y)$ is the violation rate and $E[v(X, Y)]$ is its expected violation rate for independent items. The violation rate is defined as the fraction of transactions that contain some, but not all, of the items. Collective strength gives 0 for perfectly negatively correlated items, infinity for perfectly positively correlated items, and 1 if the items co-occur as expected under independence.

For items with medium to low probabilities, the expected violation rate can be dominated by the proportion of transactions that contain neither item. In that case, collective strength produces values close to one, even if the itemset appears several times more often than expected to occur together.

Range: $[0, \infty)$

Confidence Boost

Reference: Balcázar (2013)

Confidence boost is the ratio of a rule's confidence to the highest confidence of any more general rule (i.e., a rule with the same consequent and one or more items removed from the LHS).

$$\text{confidence-boost}(X \Rightarrow Y) = \frac{\text{conf}(X \Rightarrow Y)}{\max_{X' \subset X} \text{conf}(X' \Rightarrow Y)} = \frac{\text{conf}(X \Rightarrow Y)}{\text{conf}(X \Rightarrow Y) - \text{improvement}(X \Rightarrow Y)}$$

Values larger than 1 mean the new rule boosts the confidence compared to the best, more general rule. The measure is related to the improvement measure.

Range: $[0, \infty)$ (> 1 indicates a rule with confidence boost)

Conviction

Reference: Brin et al. (1997)

$$\text{conviction}(X \Rightarrow Y) = \frac{1 - \text{supp}(Y)}{1 - \text{conf}(X \Rightarrow Y)} = \frac{P(X)P(\bar{Y})}{P(X \cap \bar{Y})}$$

where $\bar{Y} = E_{\neg Y}$ is the event that Y does not appear in a transaction. Conviction was developed as an alternative to confidence, which does not adequately capture the direction of associations on its own. Conviction compares the probability that X appears without Y under independence with the observed frequency of X without Y . In that respect, it is similar to lift (see the section about lift on this page). However, in contrast to lift, it is a directed measure since it also uses the information of the absence of the consequent. An interesting fact is that conviction is monotone in confidence and lift.

Range: $[0, \infty)$ (1 indicates independence; rules that always hold have ∞)

Cosine

Reference: Tan, Kumar, and Srivastava (2004)

Cosine is a null-invariant measure of correlation between the items in X and Y defined as

$$\text{cosine}(X \Rightarrow Y) = \frac{\text{supp}(X \cup Y)}{\sqrt{(\text{supp}(X)\text{supp}(Y))}} = \frac{P(X \cap Y)}{\sqrt{P(X)P(Y)}} = \sqrt{P(X|Y)P(Y|X)}$$

Range: $[0, 1]$ (1 means that the two transaction sets are identical; independence does not correspond to a fixed cosine value)

Coverage

Alias: LHS Support

It measures the probability that a rule $X \Rightarrow Y$ applies to a randomly selected transaction. It is estimated by the proportion of transactions that contain the antecedent of the rule $X \Rightarrow Y$. Therefore, coverage is sometimes called antecedent support or LHS support.

$$\text{cover}(X \Rightarrow Y) = \text{supp}(X) = P(X)$$

Range: $[0, 1]$

Descriptive Confirmed Confidence

Reference: Tan, Kumar, and Srivastava (2004)

Descriptive confirmed confidence contrasts the rule's confidence with the confidence of its negated consequent.

$$\text{confirmed-conf} = \text{conf}(X \Rightarrow Y) - \text{conf}(X \Rightarrow \bar{Y}) = P(Y|X) - P(\bar{Y}|X)$$

Range: $[-1, 1]$

Difference of Confidence

Alias: DOC, Difference of Proportions

Reference: Hofmann and Wilhelm (2001)

The difference of confidence is the difference between the proportions of transactions containing Y in the two groups of transactions that do and do not contain X . For the analysis of 2×2 contingency tables, this measure of the relationship between two binary variables is typically called the difference of proportions. It is defined as

$$\text{doc}(X \Rightarrow Y) = \text{conf}(X \Rightarrow Y) - \text{conf}(\bar{X} \Rightarrow Y) = P(Y|X) - P(Y|\bar{X}) = n_{XY}/n_X - n_{\bar{X}Y}/n_{\bar{X}}$$

Range: $[-1, 1]$ (0 means statistical independence)

Example and Counter-Example Rate

This measure contrasts the example and counterexample rates.

It is defined as

$$\text{ecr}(X \Rightarrow Y) = \frac{n_{XY} - n_{\bar{X}\bar{Y}}}{n_{XY}} = \frac{P(X \cap Y) - P(X \cap \bar{Y})}{P(X \cap Y)} = 1 - \frac{1}{\text{sebag}(X \Rightarrow Y)}$$

The measure is related to the Sebag-Schoenauer Measure.

Range: $(-\infty, 1]$

Fisher's Exact Test

Reference: Hahsler and Hornik (2007)

If X and Y are independent, then n_{XY} is a realization of the random variable C_{XY} which has a hypergeometric distribution with n_Y draws from a population with n_X successes and $n_{\bar{X}}$ failures. The p-value for Fisher's one-sided exact test giving the probability of observing a contingency table with a count of at least n_{XY} given the observed marginal counts is

$$\text{p-value} = P(C_{XY} \geq n_{XY})$$

The p-value is related to hyper-confidence. Compared to the chi-squared test, Fisher's exact test also applies when cells have low expected counts. Note that each rule represents a statistical test, and correction for multiple comparisons may be necessary.

Range: $[0, 1]$ (p-value scale)

Generalized Improvement

Reference: Hahsler et al. (2023)

This measure generalizes improvement to arbitrary interest measures.

$$\text{generalizedImprovement}(X \Rightarrow Y) = \min_{X' \subset X} [M(X \Rightarrow Y) - M(X' \Rightarrow Y)]$$

where M can be any measure that increases with interestingness. The original definition of improvement uses the measure confidence.

Range: $(-\infty, \infty)$ (the actual range depends on the measure)

Generalized Increase Ratio

Reference: Hahsler et al. (2023)

This measure generalizes lift increase to arbitrary interest measures.

$$\text{INC}(X \Rightarrow Y) = \min_{X' \subset X} \left[\frac{M(X \Rightarrow Y)}{M(X' \Rightarrow Y)} \right]$$

where M can be any positive interest measure. The original definition of lift increase uses lift.

Range: $[0, \infty)$ (> 1 means an increase)

Gini Index

Reference: Tan, Kumar, and Srivastava (2004)

The Gini index measures quadratic entropy as

$$\text{gini}(X \Rightarrow Y) = P(X)[P(Y|X)^2 + P(\bar{Y}|X)^2] + P(\bar{X})[P(Y|\bar{X})^2 + P(\bar{Y}|\bar{X})^2] - P(Y)^2 - P(\bar{Y})^2$$

Range: $[0, 1/2]$ (0 means that the rule provides no information about the data set)

Hyper-Confidence

Reference: Hahsler and Hornik (2007)

Hyper-confidence measures unexpectedly high or low co-occurrence counts under the hypergeometric model. Under independence, the co-occurrence count is represented by a random variable C_{XY} whose distribution is determined by n , n_X , and n_Y . Hyper-confidence for unexpectedly high co-occurrence is

$$\text{hyper-conf}(X \Rightarrow Y) = 1 - P(C_{XY} \geq n_{XY} \mid n, n_X, n_Y)$$

A confidence level greater than 0.95 indicates that the probability of observing a count at least this large under independence is less than 5%. For positive associations, hyper-confidence is one minus the one-sided p-value from Fisher's exact test. Note that each rule represents a statistical test and correction for multiple comparisons may be necessary.

Hyper-confidence can also be used to detect substitutes, for which the observed co-occurrence count is unexpectedly low:

$$\text{hyper-conf}_{\text{substitute}}(X \Rightarrow Y) = 1 - P(C_{XY} \leq n_{XY} \mid n, n_X, n_Y)$$

Range: $[0, 1]$

Hyper-Lift

Reference: Hahsler and Hornik (2007)

Hyper-lift adapts lift by replacing the expected count under independence, $E[C_{XY}] = n_X n_Y / n$, with a high quantile of the hypergeometric count distribution. This is more robust for low counts and results in fewer false positives when hyper-lift is used for rule filtering. Hyper-lift is defined as:

$$\text{hyper-lift}_\delta(X \Rightarrow Y) = \frac{n_{XY}}{Q_\delta[C_{XY}]}$$

where n_{XY} is the number of transactions containing X and Y and $Q_\delta[C_{XY}]$ is the δ -quantile of the hypergeometric distribution determined by n , n_X , and n_Y . The value of δ is typically chosen as 0.99 or 0.95.

Range: $[0, \infty)$ (a value greater than 1 means that the observed count exceeds the selected quantile)

Imbalance Ratio

Alias: IR

Reference: Wu, Chen, and Han (2010)

The imbalance ratio measures the difference in the marginal frequencies of the LHS and RHS. The ratio is close to 0 if the conditional probabilities are similar (i.e., very balanced) and close to 1 if they are very different. It is defined as

$$\text{IB}(X \Rightarrow Y) = \frac{|P(X|Y) - P(Y|X)|}{P(X|Y) + P(Y|X) - P(X|Y)P(Y|X)} = \frac{|supp(X) - supp(Y)|}{supp(X) + supp(Y) - supp(X \cup Y)}$$

Range: $[0, 1]$ (0 indicates balanced marginal frequencies)

Implication Index

Reference: Gras et al. (1996)

A variation of the Lerman similarity defined as

$$\text{gras}(X \Rightarrow Y) = \sqrt{n} \frac{supp(X \cup \bar{Y}) - supp(X)supp(\bar{Y})}{\sqrt{supp(X)supp(\bar{Y})}}$$

Lower values indicate fewer counterexamples than expected under independence and therefore stronger implication.

Range: $(-\infty, \infty)$

Importance

Reference: MS Analysis Services: Microsoft Association Algorithm Technical Reference.

In the Microsoft Association Algorithm Technical Reference, confidence is called “probability,” and a measure called importance is defined as the log-likelihood of the right-hand side of the rule, given the left-hand side of the rule:

$$\text{importance}(X \Rightarrow Y) = \log_{10} \left(\frac{L(X \Rightarrow Y)}{L(\bar{X} \Rightarrow Y)} \right)$$

where L is the Laplace corrected confidence.

Range: $(-\infty, \infty)$

Improvement

Reference: Bayardo, Agrawal, and Gunopulos (2000)

The improvement of a rule is the minimum difference between its confidence and the confidence of any proper sub-rule with the same consequent. A large positive value indicates that the more specific rule (with an additional item in the LHS) improves the confidence and should be kept. Improvement is often used to filter redundant rules.

$$\text{improvement}(X \Rightarrow Y) = \min_{X' \subset X} [\text{conf}(X \Rightarrow Y) - \text{conf}(X' \Rightarrow Y)]$$

Range: $[-1, 1]$

Jaccard Coefficient

Reference: Tan, Kumar, and Srivastava (2004)

A null-invariant measure for dependence using the Jaccard similarity index between the two sets of transactions that contain the items in X and Y , respectively. Defined as

$$\text{jaccard}(X \Rightarrow Y) = \frac{\text{supp}(X \cup Y)}{\text{supp}(X) + \text{supp}(Y) - \text{supp}(X \cup Y)} = \frac{P(X \cap Y)}{P(X) + P(Y) - P(X \cap Y)}$$

Range: $[0, 1]$

J-Measure

Reference: Smyth and Goodman (1991)

The J-measure is a scaled version of cross entropy to measure the information content of a rule.

$$J(X \Rightarrow Y) = P(X \cap Y) \log \left(\frac{P(Y|X)}{P(Y)} \right) + P(X \cap \bar{Y}) \log \left(\frac{P(\bar{Y}|X)}{P(\bar{Y})} \right)$$

Range: $[0, 1/e]$ when natural logarithms are used (0 means that X provides no information about Y)

Kappa

Alias: Cohen's κ

Reference: Tan, Kumar, and Srivastava (2004)

For a rule viewed as a classifier, Cohen's kappa coefficient adjusts the observed accuracy, $P(X \cap Y) + P(\bar{X} \cap \bar{Y})$, by the accuracy expected under independence. Kappa is defined as

$$\kappa(X \Rightarrow Y) = \frac{P(X \cap Y) + P(\bar{X} \cap \bar{Y}) - P(X)P(Y) - P(\bar{X})P(\bar{Y})}{1 - P(X)P(Y) - P(\bar{X})P(\bar{Y})}$$

Range: $[-1, 1]$ (0 means the rule is not better than a random classifier)

Klösgen

Reference: Tan, Kumar, and Srivastava (2004)

Defined as a scaled version of the added value measure.

$$\begin{aligned} \text{klosgen}(X \Rightarrow Y) &= \sqrt{\text{supp}(X \cup Y)} (\text{conf}(X \Rightarrow Y) - \text{supp}(Y)) \\ &= \sqrt{P(X \cap Y)} (P(Y|X) - P(Y)) \\ &= \sqrt{P(X \cap Y)} AV(X \Rightarrow Y) \end{aligned}$$

Range: $[-1, 1]$ (0 for independence)

Kulczynski

Reference: Wu, Chen, and Han (2010)

The null-invariant Kulczynski measure averages confidence in both directions.

$$\begin{aligned} \text{kulc}(X \Rightarrow Y) &= \frac{1}{2} (\text{conf}(X \Rightarrow Y) + \text{conf}(Y \Rightarrow X)) = \frac{1}{2} \left(\frac{\text{supp}(X \cup Y)}{\text{supp}(X)} + \frac{\text{supp}(X \cup Y)}{\text{supp}(Y)} \right) \\ &= \frac{1}{2} (P(X|Y) + P(Y|X)) \end{aligned}$$

Range: $[0, 1]$ (independence does not correspond to a fixed value)

Lambda

Alias: Goodman-Kruskal's λ , Predictive Association

Reference: Tan, Kumar, and Srivastava (2004)

Goodman and Kruskal's lambda assesses the association between the LHS and RHS of the rule.

$$\lambda(X \Rightarrow Y) = \frac{\max(n_{XY}, n_{X\bar{Y}}) + \max(n_{\bar{X}Y}, n_{\bar{X}\bar{Y}}) - \max(n_Y, n_{\bar{Y}})}{n - \max(n_Y, n_{\bar{Y}})}$$

Range: $[0, 1]$

Laplace Corrected Confidence

Alias: Laplace Accuracy, L

Reference: Tan, Kumar, and Srivastava (2004)

$$L(X \Rightarrow Y) = \frac{n_{XY} + 1}{n_X + k},$$

where k is the number of classes in the domain. For association rules, k is often set to 2. It is an approximate measure of the expected rule accuracy, representing one minus the Laplace expected error estimate of the rule. The Laplace-corrected accuracy estimate decreases with lower support to account for estimation uncertainty with low counts.

Range: $[0, 1]$

Least Contradiction

Reference: Azé and Kodratoff (2002)

$$\text{least-contradiction}(X \Rightarrow Y) = \frac{\text{supp}(X \cup Y) - \text{supp}(X \cup \bar{Y})}{\text{supp}(Y)} = \frac{P(X \cap Y) - P(X \cap \bar{Y})}{P(Y)}$$

Range: $(-\infty, 1]$

Lerman Similarity

Reference: Lerman, I.C. (1981). *Classification et analyse ordinale des données*. Paris.

Defined as

$$\text{lerman}(X \Rightarrow Y) = \frac{n_{XY} - \frac{n_X n_Y}{n}}{\sqrt{\frac{n_X n_Y}{n}}} = \sqrt{n} \frac{\text{supp}(X \cup Y) - \text{supp}(X)\text{supp}(Y)}{\sqrt{\text{supp}(X)\text{supp}(Y)}}$$

Range: $(-\infty, \infty)$

Leverage

Alias: Piatetsky-Shapiro, PS

Reference: Piatetsky-Shapiro (1991)

$$\text{PS}(X \Rightarrow Y) = \text{leverage}(X \Rightarrow Y) = \text{supp}(X \Rightarrow Y) - \text{supp}(X)\text{supp}(Y) = P(X \cap Y) - P(X)P(Y)$$

Leverage measures the difference of X and Y appearing together in the data set and what would be expected if X and Y were statistically independent. In a sales setting, it describes how many more (or fewer) baskets contain both X and Y than expected under independence.

Using minimum leverage thresholds incorporates at the same time an implicit frequency constraint. For example, with a minimum leverage threshold of 0.01% (corresponding to 10 occurrences in a data set with 100,000 transactions), one can first find all itemsets with minimum support of 0.01% and then filter the resulting itemsets using the leverage constraint. Because of this property, leverage also can suffer from the rare item problem.

Leverage is an unnormalized version of the phi correlation coefficient.

Range: $[-1/4, 1/4]$ (0 indicates independence)

Lift

Alias: Interest, interest factor

Reference: Brin et al. (1997)

Lift was originally called interest by Brin et al. Later, lift, the name of an equivalent measure popular in advertising and predictive modeling became more common. Lift is defined as

$$\text{lift}(X \Rightarrow Y) = \text{lift}(Y \Rightarrow X) = \frac{\text{conf}(X \Rightarrow Y)}{\text{supp}(Y)} = \frac{P(Y|X)}{P(Y)} = \frac{P(X \cap Y)}{P(X)P(Y)} = n \frac{n_{XY}}{n_X n_Y}$$

Lift measures how many times more often X and Y occur together than expected if they were statistically independent. A lift value of 1 indicates independence between X and Y . For statistical tests, see the chi-squared test statistic, Fisher's exact test, and hyper-confidence.

Lift is not downward closed and does not suffer from the rare item problem. However, lift is susceptible to noise in small databases. Rare itemsets with low counts (low probability), which by chance occur a few times (or only once) together, can produce enormous lift values.

Range: $[0, \infty)$ (1 means independence)

Lift Increase

Reference: López et al. (2014)

Lift increase is related to improvement, but uses lift. It divides a rule's lift by the largest lift of any proper subrule with the same consequent.

$$\text{LIC}(X \Rightarrow Y) = \min_{X' \subset X} \left[\frac{\text{lift}(X \Rightarrow Y)}{\text{lift}(X' \Rightarrow Y)} \right]$$

López et al. (2014) suggests that rules should satisfy $\text{LIC} > 1.05$ to justify adding an item to the antecedent.

Range: $[0, \infty)$ (> 1 means an increase)

Max-Confidence

Reference: Tan, Kumar, and Srivastava (2004)

Max-confidence is a symmetric, null-invariant version of confidence:

$$\text{maxConf}(X \Rightarrow Y) = \max\{\text{conf}(X \Rightarrow Y), \text{conf}(Y \Rightarrow X)\} = \max\{P(Y|X), P(X|Y)\}$$

Range: $[0, 1]$

Mutual Information

Alias: Uncertainty

Reference: Tan, Kumar, and Srivastava (2004)

Mutual information measures the information obtained about Y by observing X .

$$\begin{aligned}
M(X \Rightarrow Y) &= \frac{\sum_{i \in \{X, \bar{X}\}} \sum_{j \in \{Y, \bar{Y}\}} \frac{n_{ij}}{n} \log \frac{n n_{ij}}{n_i n_j}}{\min \left(- \sum_{i \in \{X, \bar{X}\}} \frac{n_i}{n} \log \frac{n_i}{n}, - \sum_{j \in \{Y, \bar{Y}\}} \frac{n_j}{n} \log \frac{n_j}{n} \right)} \\
&= \frac{\sum_{i \in \{X, \bar{X}\}} \sum_{j \in \{Y, \bar{Y}\}} P(i \cap j) \log \frac{P(i \cap j)}{P(i)P(j)}}{\min \left(- \sum_{i \in \{X, \bar{X}\}} P(i) \log P(i), - \sum_{j \in \{Y, \bar{Y}\}} P(j) \log P(j) \right)}
\end{aligned}$$

Range: $[0, 1]$ (0 means that X provides no information about Y)

Netconf

Reference: Tan, Kumar, and Srivastava (2004)

Netconf normalizes leverage by the maximum possible positive deviation for the antecedent:

$$netconf(X \Rightarrow Y) = \frac{leverage(X \Rightarrow Y)}{P(X)(1 - P(X))} = \frac{P(X \cap Y) - P(X)P(Y)}{P(X)(1 - P(X))}.$$

Range: $[-1, 1]$ (0 for independence)

Odds Ratio

Reference: Tan, Kumar, and Srivastava (2004)

For the analysis of 2×2 contingency tables, the odds ratio is a measure of the relationship between two binary variables. It is defined as the ratio of the odds of a transaction containing Y in the groups of transactions that contain and do not contain X .

$$OR(X \Rightarrow Y) = \frac{\frac{P(Y|X)}{1-P(Y|\bar{X})}}{\frac{P(Y|\bar{X})}{1-P(Y|X)}} = \frac{\frac{conf(X \Rightarrow Y)}{1-conf(X \Rightarrow Y)}}{\frac{conf(\bar{X} \Rightarrow Y)}{1-conf(\bar{X} \Rightarrow Y)}} = \frac{n_{XY}n_{\bar{X}\bar{Y}}}{n_{X\bar{Y}}n_{\bar{X}Y}}$$

A confidence interval around the odds ratio can be calculated (Li et al. 2014) using a normal approximation.

$$\omega = z_{1-\alpha/2} \sqrt{\frac{1}{n_{XY}} + \frac{1}{n_{X\bar{Y}}} + \frac{1}{n_{\bar{X}Y}} + \frac{1}{n_{\bar{X}\bar{Y}}}}$$

$$CI(X \Rightarrow Y) = [OR(X \Rightarrow Y) \exp(-\omega), OR(X \Rightarrow Y) \exp(\omega)]$$

where $z_{1-\alpha/2}$ is the standard normal critical value for a confidence level of $1 - \alpha$.

Range: $[0, \infty)$ (1 indicates that Y is not associated with X)

Phi Correlation Coefficient

Reference: Tan, Kumar, and Srivastava (2004)

The Phi correlation coefficient between the transactions containing X and Y , represented as two binary vectors. Phi correlation is equivalent to Pearson's product-moment correlation coefficient ρ for 0–1 values and is related to the chi-squared test statistic for 2×2 contingency tables.

$$\phi(X \Rightarrow Y) = \frac{nn_{XY} - n_X n_Y}{\sqrt{n_X n_Y n_{\bar{X}} n_{\bar{Y}}}} = \frac{P(X \cap Y) - P(X)P(Y)}{\sqrt{P(X)(1 - P(X))P(Y)(1 - P(Y))}}$$

In machine learning, phi correlation is also known as the Matthews correlation coefficient (MCC). The magnitude of the correlation is also related to the chi-squared statistic:

$$|\phi(X \Rightarrow Y)| = \sqrt{\frac{\chi^2}{n}}$$

Range: $[-1, 1]$ (0 when X and Y are independent)

Ralambondrainy

Reference: Diatta, Ralambondrainy, and Totohasina (2007)

This measure is the support of the counterexamples.

$$\text{ralambondrainy}(X \Rightarrow Y) = \frac{n_{X\bar{Y}}}{n} = \text{supp}(X \cup \bar{Y}) = P(X \cap \bar{Y})$$

Range: $[0, 1]$ (smaller is better)

Relative Linkage Disequilibrium

Reference: Kenett and Salini (2008)

RLD is an association measure motivated by indices used in population genetics. It evaluates the deviation of the rule's support from the support expected under independence, given the supports of X and Y .

$$D = \frac{n_{XY}n_{\bar{X}\bar{Y}} - n_{X\bar{Y}}n_{\bar{X}Y}}{n}$$

$$\text{RLD} = \begin{cases} D/(D + \min(n_{X\bar{Y}}, n_{\bar{X}Y})) & \text{if } D > 0, \\ D/(D - \min(n_{XY}, n_{\bar{X}\bar{Y}})) & \text{otherwise.} \end{cases}$$

Range: $[0, 1]$

Relative Risk

Reference: Siström and Garvan (2004)

For the analysis of 2×2 contingency tables, relative risk is a measure of the relationship between two binary variables. It is the ratio of the proportions of transactions containing Y in the two groups of transactions that contain and do not contain X . In epidemiology, this corresponds to the ratio of the risk of having disease Y in the exposed (X) and unexposed ($\bar{X}$) groups.

$$RR(X \Rightarrow Y) = \frac{n_{XY}/n_X}{n_{\bar{X}Y}/n_{\bar{X}}} = \frac{P(Y|X)}{P(Y|\bar{X})} = \frac{conf(X \Rightarrow Y)}{conf(\bar{X} \Rightarrow Y)}$$

Range: $[0, \infty)$ ($RR = 1$ means X and Y are unrelated)

Rule Power Factor

Reference: Ochin and Kumar (2016)

The rule power factor weights a rule's confidence by its support and therefore favors rules that have both high confidence and high support.

It is defined as

$$rpf(X \Rightarrow Y) = supp(X \Rightarrow Y) \cdot conf(X \Rightarrow Y) = \frac{P(X \cap Y)^2}{P(X)}$$

Range: $[0, 1]$

Right-Hand-Side Support

Alias: RHS support, consequent support

Support of the right-hand side of the rule.

$$RHSsupp(X \Rightarrow Y) = supp(Y) = P(Y)$$

Range: $[0, 1]$

Sebag-Schoenauer

Reference: Sebag and Schoenauer (1988)

It is defined as

$$sebag(X \Rightarrow Y) = \frac{conf(X \Rightarrow Y)}{conf(X \Rightarrow \bar{Y})} = \frac{P(Y|X)}{P(\bar{Y}|X)} = \frac{supp(X \cup Y)}{supp(X \cup \bar{Y})} = \frac{P(X \cap Y)}{P(X \cap \bar{Y})}$$

Range: $[0, \infty)$

Standardized Lift

Reference: McNicholas, Murphy, and O'Regan (2008)

Standardized lift uses the minimum and maximum values that lift can attain for each rule to map lift to the interval from 0 to 1. Its lower bound is

$$\lambda = \frac{\max\{P(X) + P(Y) - 1, 1/n\}}{P(X)P(Y)}.$$

Its upper bound is

$$v = \frac{1}{\max\{P(X), P(Y)\}}$$

The standardized lift is defined as

$$\text{stdLift}(X \Rightarrow Y) = \frac{\text{lift}(X \Rightarrow Y) - \lambda}{v - \lambda}.$$

The standardized lift measure can account for the minimum support s and minimum confidence c used in rule mining by replacing the lower bound λ with

$$\lambda^* = \max \left\{ \lambda, \frac{4s}{(1+s)^2}, \frac{s}{P(X)P(Y)}, \frac{c}{P(Y)} \right\}.$$

Range: $[0, 1]$

Varying Rates Liaison

Reference: Bernard and Charron (1996)

Varying rates liaison is defined as the lift of a rule minus 1; therefore, 0 represents independence.

$$\text{VRL}(X \Rightarrow Y) = \text{lift}(X \Rightarrow Y) - 1$$

Range: $[-1, \infty)$ (0 for independence)

Yule's Q

Reference: Tan, Kumar, and Srivastava (2004)

Yule's Q, also called Yule's coefficient of association, is a special case of the Goodman and Kruskal's gamma. It is defined as

$$Q(X \Rightarrow Y) = \frac{\alpha - 1}{\alpha + 1}$$

where $\alpha = OR(X \Rightarrow Y)$ is the odds ratio of the rule.

Range: $[-1, 1]$

Yule's Y

Reference: Tan, Kumar, and Srivastava (2004)

Yule's Y is also known as the coefficient of colligation to measure the association between two binary variables. It is defined as

$$Y(X \Rightarrow Y) = \frac{\sqrt{\alpha} - 1}{\sqrt{\alpha} + 1}$$

where $\alpha = OR(X \Rightarrow Y)$ is the odds ratio of the rule.

Range: $[-1, 1]$

Zhang's Measure

Reference: Tan, Kumar, and Srivastava (2004)

Zhang's measure is a bounded dependence score that can be viewed as a scaled version of leverage:

$$Z(X \Rightarrow Y) = \frac{leverage(X \Rightarrow Y)}{\max\{P(X \cap Y)(1 - P(X)), P(X)(P(Y) - P(X \cap Y))\}} = \frac{P(X \cap Y) - P(X)P(Y)}{\max\{P(X \cap Y)(1 - P(X)), P(X)(P(Y) - P(X \cap Y))\}}$$

Range: $[-1, 1]$ (0 for independence)

Measures for Classification Rules

For rules interpreted as binary classifiers, the following measures are available. They are computed from the contingency-table counts as follows.

The count variables use the first subscript for the rule antecedent X and the second subscript for the consequent Y :

	Y	$\bar{Y}$	Total
X	n_{XY}	$n_{X\bar{Y}}$	n_X
$\bar{X}$	$n_{\bar{X}Y}$	$n_{\bar{X}\bar{Y}}$	$n_{\bar{X}}$
Total	n_Y	$n_{\bar{Y}}$	n

Accuracy

Accuracy is the proportion of transactions classified correctly by the rule, including both positive and negative predictions.

$$accuracy = (n_{XY} + n_{\bar{X}\bar{Y}})/n.$$

Range: $[0, 1]$

Precision

Precision is the proportion of transactions predicted positive by the rule that are actually positive.

$$precision = n_{XY} / n_X.$$

Range: $[0, 1]$

Recall

Recall is the proportion of actual positive transactions that are correctly identified by the rule.

$$recall = n_{XY} / n_Y.$$

Range: $[0, 1]$

F-score

The F-score is the harmonic mean of precision and recall, balancing the two types of classification performance.

$$F_1 = 2n_{XY} / (n_X + n_Y).$$

Range: $[0, 1]$

Balanced Accuracy

Balanced accuracy is the mean of the true-positive and true-negative rates, giving equal weight to the positive and negative classes.

$$balancedAccuracy = \frac{1}{2}(n_{XY} / n_Y + n_{\overline{X}\overline{Y}} / n_{\overline{Y}}).$$

Range: $[0, 1]$

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